Real Time Face Recognition based on Convolution Neural Network

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Abstract—Face recognition is an active research area with many successful traditional and deep learning methods. In our research, we are introducing a Real Time Face Recognition based on Convolution Neural Network (CNN) method for face recognition and to maintain a high accuracy which is running in real time.

Face represents multidimensional, complex, meaningful visualization and developing a real time face recognition computational model for face recognition is not easy.

In present time, real time face recognition is a challenging problem. The main challenge of face recognition is that how to improve the image recognition performance of the system when it is affected by the variability of non-linear effects. In this paper, 3-layer CNN architecture is proposed for solving the face recognition problem, CNN gives a solution which is capable of solving the facial images related problems that contains varying illumination, poses, occlusions and facial expressions.

The test on 115 subject of Essex University database achieve an accuracy of 98.5% which is similar range of performance as the best result of previous works. Meaningfully, our system of real time face recognition based on CNN completes the facial recognition process in less than 0.019 seconds.

Keywords: Convolution Neural Network, RELU

I. INTRODUCTION

Face recognition is an old concept about human-computer interaction. This concept was proposed in 1960s, there have been about five to six methods implemented for face recognition, which is including the subspace analysis method [1], the hidden Markov model method [2], the geometrical characteristic method [3], the neural network method [4] and the elastic graph matching method [5]. In this, first four algorithms are about extracting simple feature and classifying but the difficult features of face are extracted by Neural Network by using deep learning so that more accuracy is obtained. In face verification, Sun Y [11, 12, 13] researched and developed Deep Id method based on CNN.

Generally, face recognition [6] is divided into two subcategories. On the one side, face identification (or 1:1 face recognition) consist in finding the identity corresponding to a given face. On the other side, face verification (or 1:1 face recognition) consists on checking if a face–corresponds to a given identity. Real time face recognition can be separated in terms of evaluation protocol. "Algorithms are tested under the same as the training ones and face recognition can then be assimilated to classification problem. In the testing case, usually the identities of testing set is separated from the training set. The problem becomes more one of encoding face in to a discriminative feature space."

For personal identification of any person through computerized access can be done by our face recognition system, it's resulted in an increased interest in biometric. Biometric being investigated include face, fingerprint [7], Speech [8], and signature dynamics recognition [9].

Face recognition is also known as a passive non–intrisive system because it verifies personal identity. The technique used in face recognition system may depend on the application of the system.

II. SYSTEM MODEL DESIGN

A. *The Structure of System*

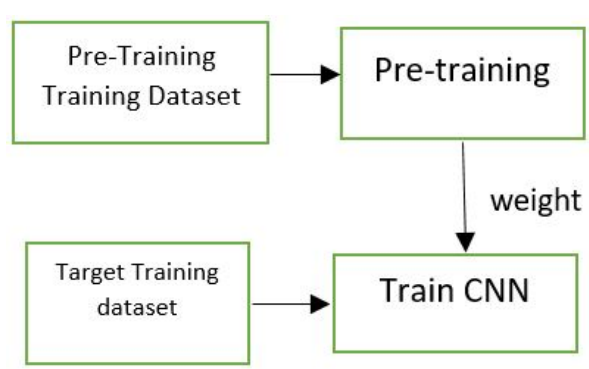

Figure 1: Training framework of system

In our system, all images are scaled to 36 x 40 Resolution and converted to grayscale from BGR to feed to our network. We used Our face dataset as a pre-training

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dataset, which is fed to our system with random weight and after the training the weight is shared to the model to train the target dataset this helps us to decrease the training time of the network. In short, the pre–training is a method of reducing the training time by precomputed weights .

Figure 2: Testing framework of system

1) Convolution and Pooling

CNN Model in deep learning, a series of convolution layers is used , we pass images through these layers with filters(kernel),perform Pooling , and use fully connected layers and Softmax function to classify object depending on the probability.

Convolution layer: First layer is used to extract the features. In mathematical form we can represent it as:

- Input image matrix (h x w x d)
- A filter (f_h x f_w x d)
- output of this $((h-f_h+1) \times (w-f_w+1) \times 1)$

ReLU: ReLU is abbreviation of Rectified Linear Unit which use for a non-linearity. The output is

$f(x) = max (0, x)$.

Pooling layer: Pooling layer is used to reduce the feature parameters when we use large images. Here we have used Max Pooling, that take max element from the feature Map.

Fully Connected layers: in this layer we flatten the matrix into a vector to feed into a fully connected layer.

2) CNN Model Design

In this, we are introducing a CNN [10] based research paper. In above figure you have seen that a CNN consists of an input, an output layer and multiple hidden layers. There are many other layers which are present in the hidden layers of CNN such as Pooling layers, Convolution layers, normalization layers, and fully connected layers. In neural networks, the explanation of each process as a convolution is done by convention.

CNN is proposed by Yann LeCun in 1988. CNN is a special architecture of artificial neural networks.

Most popular uses of CNN is image classification and for analyzing visual imagery. For example Automatic tagging algorithm in Facebook is done by CNN. Let us consider the use of CNN for image.

The main task of image classification is acceptance of the image and giving the specific definition of its class. This is same as a learning process of human being that learn from their birth and after few year they are able to easily identify that the image in the picture is a human or not. But the computer reads and sees the pictures quite different from the human:

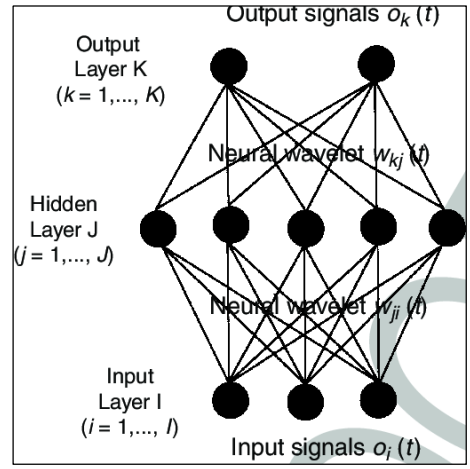
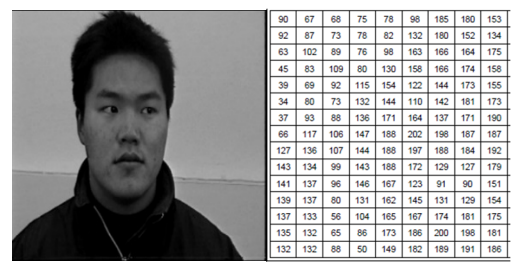


Figure 3: Three layer CNN architecture



90	67	68	75	78	98	185	180	153
92	87	73	78	82	132	180	152	134
63	102	89	76	98	163	166	164	175
45	83	109	80	130	158	166	174	158
39	69	92	115	154	122	144	173	155
34	80	73	132	144	110	142	181	173
37	93	88	136	171	164	137	171	190
66	117	106	147	188	202	198	187	187
127	136	107	144	188	197	188	184	192
143	134	99	143	188	172	129	127	179
141	137	96	146	167	123	91	90	151
139	137	80	131	162	145	131	129	154
137	133	56	104	165	167	174	181	175
135	132	65	86	173	186	200	198	181
132	132	88	50	149	182	189	191	186

Figure 4: Image read by computer

In the above fig. 3 the computer sees only an array of pixel instead of the image. For example, see the above image, if the image size is 200 x 200 then size of the array will be 200 x 200 x 3. Where first 200 is width of image, next 200 is height of image and 3 is RGB channel values. This RGB channel value is assigned by a computer to each of these pixels from 0 to 255.

Instead of the image, the computer sees an array of pixels. For example, if image size is 200 x 200. In this case, the size of the array will be 200x200x3. Where 200 is width, next 200 is height and 3 is RGB channel values. The computer is assigned a value from 0 to 255 to each of these numbers. This value expresses the intensity of each pixel point.

For More detail see the following flow diagram:

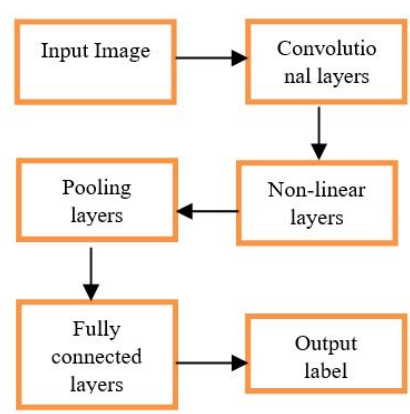


Figure 5: Convolution neural network processes

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The more recent measurements are more relevant to the task, so a weighted average data with more weighted on the recent measurement gives a better input signal. This can be done by the weighting function w(n), where n is the age of a measurement. The operation that is applied here is called the convolution operator:

$$\hat{y}(i) = \sum_n x(i-n)w(n) \quad (1)$$

Where $\hat{y}$ is the output function of the convolution operator to keep it consistent with the actual output of a neuron, x(i) is a noise input signal with measurements for time step i.

The convolution operation can be applied on more than one axis at a time. The convolution formula will be changed as follows:

$$\hat{y}(i, j) = \sum_m \sum_n x(i-m, j-n)w(m, n) \quad (2)$$

where i and j are the indexes of the two-dimensional input image and m and n are the indexes of the two-dimensional image.

3) *Pre–Training of Model*

In our research, we are focusing on mainly reducing the training time of the model when we have a large dataset. In the today's world, we have large volume of data and less time to train the data. So, we used a pretraining approach to get the weight of the nodes in convolutional neural network to use to train the target dataset. So that the number of Epoch to get maximum accuracy can be decreased, because we can get the appropriate weight in less number of epochs.

III. Datasets

Our dataset contains of two datasets:

A. *Our Face Database*

Our face database consists of faces of 90 peoples and there are 74 faces of each person of 480 x 640 resolution making it a total 6660 images, which are used for pre-training of our model to share weights with the model to train our target dataset in less time.

B. *Essex University Face Database*

Essex University face database consists of faces of 115 peoples and there are 20 faces of each person of 180*200 resolution. Each individual faces is divided into group of 14 and 6. And, Every Image is flipped horizontally to increase the number of images and complexity to the model. So we have 28 images for training data set and 12 images is for testing dataset making it the training size equal to 28 * 115 = 3220 image and 12 * 115 = 1380 images.

IV. Experimental Analysis

A. *CNN Performance Analysis*

As we trained our model with the targeted data with random weights with standard deviation of 0.1 ,the model takes averagely takes 14 epoch to get maximum accuracy ,while pre-training our model with another face database and generating the correct weight ,after that sharing that weight to the model to train our targeted data. In this research, we analyzed that pre–training and weight sharing reduces the number of epochs to get maximum training accuracy to 7 epochs, that is half of the non–pretrained model.

And performance is nearly same for both models, equal to 93.5 training accuracy and 98.6 test accuracy.

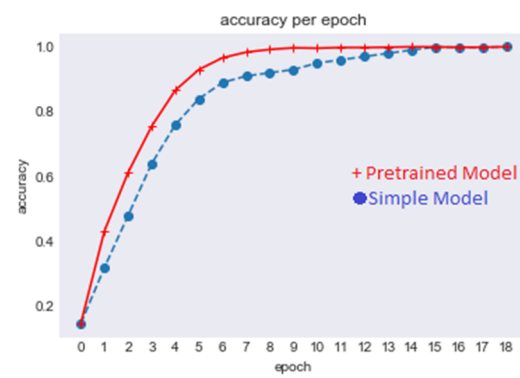

Figure 6: Accuracy per epoch graph

B. *Training Time Analysis*

Training time of the model is as follows:

Model	Training Time		Test Accuracy
Normal CNN	132 s		98.4%
Pre-Trained CNN	66 s		98.6%

As from the above the table, the training time for training the targeted data using pre-trained CNN model is half as compared to Normal CNN. Since the number of epochs is the pre-trained model is less, so the training time is also less.

V. Conclusion

In this research paper, I have worked on using pre–training the model technique to reduce the number of epoch while achieving maximum accuracy so that less training time is required for the targeted data. This experiment is based on Essex University Face database which verifies the advantage of our system. In the future, we will try to minimize the training time while maximizing accuracy with a larger set of face dataset.

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